Your Bot Is a System

Capstone review: from language to procedures to a system – what ELIZA can do so far, and one function for many rules.

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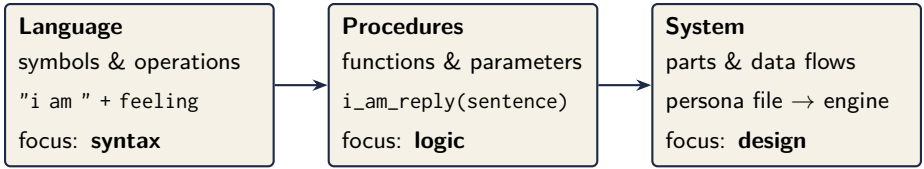
Fall 2026

Today

1. From language to system
2. Your bot is a system
3. What ELIZA can do so far
4. Three functions: what differs?
5. One function, many rules
6. From a call to a rule: what becomes data
7. Practice: a rule of your own
8. What the capstone adds

From language to system

In an AI future, AI writes more and more of the syntax and the procedures. Software engineering means: **you design the parts and their connections, and you write the tests that show it works.**

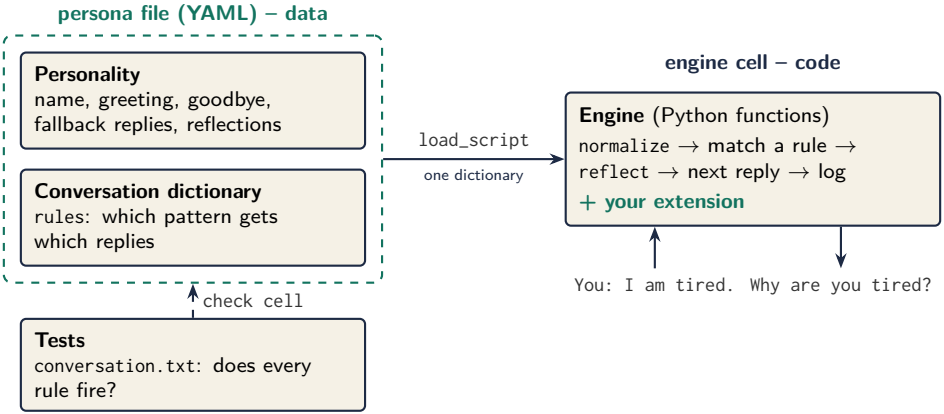


Each level is built from the one before: a system is made of procedures, and procedures are written in the language. At the system level, the bugs sit in the connections between the parts.

Software engineering: your bot is a system

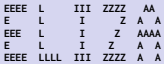
```
EEEE L    III ZZZZ  AA
E    L     I    Z  A A
EEE  L     I   Z   AAAA
E    L     I  Z    A  A
EEEE LLLL III ZZZZ A  A
```

A **system** is a set of parts, each with one job, and defined connections between them.



A new rule is **new data**, not new code: the engine stays the same.

What ELIZA can do so far



Date	Lecture	What ELIZA can do now
10/26-28	Spot a Keyword	notices keywords in your sentence: reads them from `keywords.txt`, finds them with `in` after `lower()`
10/30	The First Reply	picks the first keyword and answers with a fixed reply, or a fallback; doctests test the functions
11/04-06	Why Are You Tired?	reuses your words: "I am tired" → "Why are you tired?"; tidies up what you type, logs the conversation
11/09	Why Do You Need a Break?	one pattern per rule, `group(1)` cuts out your words; `\b` keeps "mother" out of "smothered"
next	**Can You Tell Me More?**	**rules = list items, tried top to bottom; replies take turns; words split into a list: "my" → "your"**

Colab: `birkenkrahe.github.io/go/elizareview`